

FULL FRAMEWORK V3 Independent Reproducibility Audit

Final evidence record for HIS Synthetic Benchmark V1.3, seed 42, corruption rate 5 percent
Audit completed 8 October 2026

Audit conclusion

The independent Windows reproduction passed for the tested condition. All six cleaned tables were byte-identical to the retained original outputs. The original-machine and clean-machine runs contained the same multiset of 209,311 cleaning actions after excluding sequence-generated action identifiers. All 106,698 matched injected events were attributed to identical action content. The v1.0.2 serial-attribution-corrected evaluator produced identical metrics from the two action logs.

This conclusion is deliberately bounded. The action logs and event-matching traces were not byte-identical, and the audit does not establish reproducibility for untested seeds, corruption rates, datasets, operating systems, or clinical deployment.

Scope and environment

Item	Audited setting
Benchmark	HIS Synthetic Benchmark V1.3
Cleaner	Frozen FULL_FRAMEWORK_V3
Condition	seed = 42; corruption rate = 5%
Independent environment	Windows; Python 3.10.11; isolated virtual environment
Corrected evaluator	evaluate_his_v1_3_phase4a6_final_v3_serial_attribution_fixed.py
Evaluator SHA-256	58bedfc98ca9b7a375a558d0423bd682432b165ba4907d056fac128c7a55f918

Acceptance results

Check	Observed result	Status
Independent cleaner execution	Exit code 0	PASS
Cleaned tables	6 of 6 SHA-256 identical	PASS
Action content	209,311 actions; non-ID multiset identical	PASS
Injected-event matching	160,048 event statuses identical	PASS
Matched-event attribution	106,698 of 106,698 action contents identical	PASS
Attribution mismatches	0	PASS
Corrected dual-log evaluation	All reported metrics identical	PASS

Corrected evaluation metrics

Metric	Original-machine log	Clean-machine log
TP	106,698	106,698
FP	8,716	8,716
FN	53,350	53,350
SerialFollowUpGroupsExcluded	49	49
Precision	0.9244805656	0.9244805656
Recall	0.6666625012	0.6666625012
F1	0.7746839854	0.7746839854
CorrectEdits	161,236	161,236
IncorrectEdits	1	1
EditPrecision	0.9999937979	0.9999937979
FCR	0.00000620205	0.00000620205
NER	0	0
PostCleanCellAcc	0.7172521361	0.7172521361
FlagSafety	1	1
DupRecall	0	0
RecoveryAcc	0.5052558970	0.5052558970

File-level comparison

Evaluator output	Byte-identical
evaluation_metrics.csv	Yes
evaluation_summary.json	Yes
negative_state_evaluation_trace.csv	Yes
unmatched_action_groups.csv	Yes
event_matching_trace.csv	No - action identifiers differ

Difference attribution

The two action logs have different byte-level SHA-256 values. A total of 24,996 matched action identifiers differed, all within the previously identified M1 diagnosis fuzzy-correction ordering difference. Removing the generated identifier from the comparison yielded identical action semantics. A record-by-record mapping of every successful event to its referenced action found 106,698 identical action contents and zero mismatches. Therefore, the trace difference is classified as an ordering and identifier-assignment difference, not a difference in cleaning behavior or evaluated event attribution.

Evidence integrity

Artifact	SHA-256
Returned evidence ZIP	246861b93f432e3f3f5a83d6a1a3df61f6661c1d0d3b528af1c402484e2d3af0
Original action log	96003caa7864e6345367efca2a2abea3ae03e144017926c9898f08610eefc1b4
Clean-machine action log	fd323b571c1d0fc58833f41a87fe1746161a3407a256eeca015b975d98b5f8cc
Corrected metrics CSV, both logs	db1a122c7bbc4a9e749fbaebcef825097c1552daef80a2ffddd0c63b60a038ec

Limitations

This was a fixed-condition engineering reproducibility audit, not a new performance experiment or clinical validation. It covered one seed and one corruption rate. It does not prove that action ordering is deterministic, that every generated file is byte-identical, or that other configurations will reproduce without further testing. The prior evaluator and its outputs must remain available as provenance; corrected outputs must not silently overwrite the historical record.

Final determination

Core independent reproducibility acceptance: PASS. Public-release acceptance remains a separate release-engineering gate requiring verification of the v1.0.2 archive, manifest, Git line-ending preservation, documentation, and absence of restricted data before publication.